

Angle Geometry

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1 Preliminaries

1.1 Angle Configurations

Theorem 1. Let O , I , and H be respectively the circumcentre, incentre, and orthocentre of acute triangle ABC . Then,

1. $\angle BOC = 2\angle BAC$,
2. $\angle BIC = 90^\circ + \frac{1}{2}\angle BAC$, and
3. $\angle BHC = 180^\circ - \angle BAC$.

Exercise 2. What happens when the triangle is obtuse?

Exercise 3. Given points A, B, C, D, E, F on a circle with centre O such that $BC = EF$, we know that $\angle BOC = \angle EOF$. What does this say about angles BAC and EDF ?

One particular application of the above occurs when points A, B, C and D are on a circle such that D and A are on opposite sides of line BC , and $\angle BAD = \angle DAC$. Then, $BD = DC$.

We can generalize the above via the Law of Sines: given points A, B, C on a circle with radius r , $\sin BAC = BC/2r$.

1.2 Trigonometric Formulas

Theorem 4. We have the following trigonometric formulas:

1. $\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b$,
2. $\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b$, and
3. $\cot(a \pm b) = \frac{\cot a \cot b \mp 1}{\cot a \pm \cot b}$.

Theorem 5. Reversing the first two, we can also derive the product-to-sum and sum-to-product formulas:

1. $\sin a \sin b = \frac{1}{2}(\cos(a - b) - \cos(a + b))$, and
2. $\sin a + \sin b = 2 \sin\left(\frac{a+b}{2}\right) \cos\left(\frac{a-b}{2}\right)$.

Exercise 6. Derive similar formulas for $\sin a \cos b$, $\cos a \cos b$, and $\cos a + \cos b$.

Exercise 7. Using the law of sines, prove the law of cosines: that is, given triangle ABC , we have

$$a^2 = b^2 + c^2 - 2bc \cos BAC.$$

Often times, when using the Law of Sines, it is necessary to add up angles, since the quantities you are trying to relate do not have the same angles. A common configuration is the following: given triangle ABC with $\angle BAC = \alpha$, and $\angle CBA = \beta$, we have

$$\frac{BA}{AC} = \frac{\sin ACB}{\sin CBA} = \frac{\sin(\beta + \alpha)}{\sin \beta} = \cos \beta + \sin \beta \cot \alpha.$$

2 Proofs of some theorems

2.1 Ceva and Menelaus

Theorem 8 (Ceva). *Fix a triangle ABC and a point P , and let AP, BP, CP intersect BC, CA, AB , respectively, at points D, E, F . Then*

$$\frac{BD}{DC} \frac{CE}{EA} \frac{AF}{FB} = 1.$$

Proof. By the Law of Sines on triangles ABD and DBC , we have

$$\frac{BD}{DC} = \frac{BA \sin \angle BAP}{AC \sin \angle CAP}.$$

Taking the cyclic product finishes the problem. $\square$

Theorem 9 (Menelaus). *Fix a triangle ABC and a line ℓ , and let ℓ intersect BC, CA, AB , respectively, at points D, E, F . Then*

$$\frac{BD}{DC} \frac{CE}{EA} \frac{AF}{FB} = 1.$$

Proof. By the Law of Sines on triangle DBF , we have

$$\frac{BD}{FB} = \frac{\sin \angle EDB}{\sin \angle DFA}.$$

Taking the cyclic product finishes the problem. $\square$

2.2 Pascal and Pappus

Theorem 10 (Pascal). *For points A, B, C, D, E, F on a circle, let G be the intersection of lines AB and DE , H of BC and EF , and I of CD and FA . Then G, H, I are collinear.*

Proof. We compute

$$\frac{\sin \angle AGI}{\sin \angle IGD} \bigg/ \frac{\sin \angle BGH}{\sin \angle HGE},$$

and hope that it is equal to 1, from which we can conclude.

From Law of Sines on triangles AGI and IGD , and then on triangle IFC , we have

$$\frac{\sin \angle AGI}{\sin \angle IGD} = \frac{AI \sin \angle FAB}{ID \sin \angle CDE} = \frac{\sin \angle AFC \sin \angle FAB}{\sin \angle DCF \sin \angle CDE}.$$

Similarly,

$$\frac{\sin \angle BGH}{\sin \angle HGE} = \frac{\sin \angle FCB \sin \angle ABC}{\sin \angle EFC \sin \angle DEF}.$$

These are equal since all the angles are respectively equal. $\square$

Exercise 11 (Pappus). *Same as above, except A, C, E are collinear, as are B, D, F , as opposed to being concyclic.*

3 Some example problems

3.1 Bosnia and Herzegovina 2022 TST, #1

Problem 12. *Let ABC be a triangle such that $AB = AC$ and $\angle BAC$ is obtuse. Point O is the circumcenter of triangle ABC , and M is the reflection of A in BC . Let D be an arbitrary point on line BC , such that B is in between D and C . Line DM cuts the circumcircle of ABC in E, F . Circumcircles of triangles ADE and ADF cut BC in P, Q respectively. Prove that DA is tangent to the circumcircle of triangle OPQ .*

Proof. Since M is the reflection of A with respect to BC and D is on BC , we have that $\angle ADP = \angle PDE$ and so $AP = PE = PM$. Similarly, $AQ = QF = QM$. Since $AO = OE$, we have $\triangle APO \simeq \triangle EPO$, and so OP is the angle bisector of $\angle EOA$. Since $PE = PM$, the points E, P, M, O are concyclic, whence $\angle DAP = \angle PEM = \angle POM$, and similarly $\angle DAQ = \pi - \angle QOM$. Combining these together gives $\angle PAQ = \pi - \angle POQ$, whence P, O, Q, A are concyclic, and then the equality $\angle DAP = \angle POM$ implies that DA is tangent to circle $POQA$ at A . $\square$

3.2 India EGMO 2023 TST, #6

Problem 13. Let ABC be an isosceles triangle with $AB = AC$. Suppose P, Q, R are points on segments AC, AB, BC respectively such that $AP = QB$, $\angle PBC = 90^\circ - \angle BAC$ and $RP = RQ$. Let O_1, O_2 be the circumcenters of $\triangle APQ$ and $\triangle CRP$. Prove that $BR = O_1O_2$.

Proof. We first write out all the angles. Let the circumcircles of APQ and CRP intersect again at S . Let $\angle ABC = \theta$. Then, $\angle RSP = \pi - \theta$ and $\angle QSP = 2\theta$, so $\angle QSR = \pi - \theta$. Since $\angle QSR = \angle PSR > \pi/2$, we can use SSA congruency to show that $\triangle QSR \simeq \triangle PSR$. Thus, $QS = SP$, which implies $\angle QAS = \angle SAP$ and S is on the angle bisector of $\angle BAC$. Now, $\angle AQS = \angle SPC$, and since $AQ = PC$ and $QS = SP$, we get that $\triangle QAS \simeq \triangle PCS$, and so $AS = CS$. Since S is on the angle bisector of $\angle BAC$, we get that S is the circumcentre of triangle ABC . Finally, since $\angle SBA = \pi/2 - \theta = \angle PBA$, we get that B, S, P are collinear. Finally, we note that the congruent triangles QAS and PCS imply that the circumradii of the circles is the same, and hence O_1PO_2Q is a rhombus.

Now, by the Law of Sines on triangle BSR , we get

$$BR = SR \frac{\sin \theta}{\sin(2\theta - \pi/2)},$$

and since SR is a chord on circle $OPCR$, we have

$$SR = 2r \sin(2\theta - \pi/2),$$

and finally, from the rhombus PO_1QO_2 , we have

$$O_1O_2 = 2r \sin \theta,$$

from which the result holds. $\square$

3.3 Moldova 2022 TST, #3

Now we move onto some more computational applications of angle chasing.

Problem 14. In the acute triangle ABC the point M is on the side BC . The inscribed circle of triangle ABM touches the sides BM, MA and AB in points D, E and F , and the inscribed circle of triangle ACM touches the sides CM, MA and AC in points X, Y and Z . The lines FD and ZX intersect in point H . Prove that lines AH, XY and DE are concurrent.

Proof. Let XY and DE intersect at the point K . We want to show that A, H , and K are collinear. We make this transformation since K and H are symmetric with respect to the diagram (that is, they are fixed if we swap the pairs of points (B, C) , (D, X) , (E, Y) , and (F, Z)), and thus perhaps nicer to work with.

For the point H , we compute

$$\frac{\sin \angle DAH}{\sin \angle HAX} = \frac{DH \sin \angle ADH}{HX \sin \angle HXA},$$

by the Law of Sines on triangles DAH and XAH . For the point K , we have

$$\frac{\sin \angle DAK}{\sin \angle KAX} = \frac{DK \sin \angle ADK}{KX \sin \angle KXA},$$

by the Law of Sines on triangles ADK and AXK . Our goal is to show that the right hand sides of the two above equations are equal, as this would imply the same for the left hand sides, and thus A, H , and K are collinear (why?).

Further, we see that

$$\frac{DH}{HX} = \frac{\sin \angle DXH}{\sin \angle XDH} = \frac{\cos C/2}{\cos B/2},$$

and

$$\frac{DK}{KX} = \tan(\angle AMB/2),$$

and we have removed H and K from the diagram, effectively. Continuing, we see

$$\frac{\sin \angle ADH}{\sin \angle ADK} = \frac{\sin \angle ADF}{\sin \angle ADE} = \frac{\sin \angle AFD}{\sin \angle AED} = \frac{\cos B}{\cos(\angle AMB/2)}$$

by the Law of Sines on triangles AFD and AED . Similarly,

$$\frac{\sin \angle AXH}{\sin \angle AXK} = \frac{\cos C}{\cos(\angle AMC/2)},$$

and combining the above equations finishes the problem. □

3.4 IMO 2017, #4

Problem 15. Let R and S be different points on a circle Ω such that RS is not a diameter. Let ℓ be the tangent line to Ω at R . Point T is such that S is the midpoint of the line segment RT . Point J is chosen on the shorter arc RS of Ω so that the circumcircle Γ of triangle JST intersects ℓ at two distinct points. Let A be the common point of Γ and ℓ that is closer to R . Line AJ meets Ω again at K . Prove that the line KT is tangent to Γ .

Proof. For symmetry, assume that U is an intersection of the tangent at T to circle Γ (to be determined later). Our goal is to show that U, J , and A are collinear for some choice of U , which would show that $U = K$ and the problem.

The idea behind this transformation of the problem is that everything is now symmetric: the diagram remains invariant upon interchanging the pairs (R, T) and (A, U) . We would now like to get a handle on the angles formed by the lines AJ and JU .

For notation, let $\angle TAS = \alpha$, $\angle SAR = \pi - \beta$, and $\angle ASR = \gamma$. The relation between these angles can be computed by using the Law of Sines on triangles TJS and SJR :

$$\frac{\sin(\gamma - \alpha)}{\sin \alpha} = \frac{AS}{ST} = \frac{AS}{SR} = \frac{\sin(\gamma + \pi - \beta)}{\sin \beta},$$

which simplifies to (do this computation yourself)

$$\cot \alpha + \cot \beta = 2 \cot \gamma.$$

Now, we note that $\angle ARJ = \gamma$ and $\angle ART = \beta$, so the Law of Sines on triangle ART gives us

$$\frac{\sin(\angle TAR + \beta)}{\sin(\angle TAR)} = \frac{AR}{RT},$$

and on triangle ARJ gives us

$$\frac{\sin(\angle JAR + \gamma)}{\sin(\angle JAR)} = \frac{AR}{RJ}.$$

Note that we avoid using the lengths of AT and AJ because they do not cancel as nicely, and our goal is to get rid of the point A in the diagram. Here, angles JAR and TAR are related since $\angle TAR = \gamma + \angle JAR$. Finally, we relate $\angle JAR$ to $\angle AJS$, which is an angle that will be easier to deal with via symmetry arguments:

we have $\angle AJS = \angle JAR + \gamma + \beta$. Let $\angle AJS = \phi$. Combining all of this together, we have from the above Law of Sines,

$$\frac{RJ}{RT} = \frac{RJ}{AR} \frac{AR}{RT} = \left(\frac{\sin(\angle JAR)}{\sin(\angle JAR + \gamma)} \right) \left(\frac{\sin(\angle TAR + \beta)}{\sin(\angle TAR)} \right) = \frac{\sin(\angle JAR) \sin(\angle TAR + \beta)}{\sin^2(\angle TAR)}$$

and we now use the Law of Sines on triangle JRS for the left-hand side and write the right hand side in terms of ϕ :

$$\frac{\sin \gamma}{2 \sin \beta} = \frac{\sin(\phi - \beta - \gamma) \sin(\phi)}{\sin^2(\phi - \beta)},$$

and we can now expand the right hand side to get

$$\frac{\cos(\beta + \gamma) - \cot \phi \sin(\beta + \gamma)}{\cos^2 \beta - 2 \cot \phi \sin \beta \cos \beta + \cot^2 \phi \sin^2 \beta}.$$

Dividing the numerator by $\sin \beta \sin \gamma$ and the denominator by $\sin^2 \beta$ in order to rewrite everything in terms of cotangent, we get

$$\frac{1}{2} = \frac{\cot \beta \cot \gamma - 1 - \cot \phi (\cot \beta + \cot \gamma)}{\cot^2 \beta - 2 \cot \phi \cot \beta + \cot^2 \phi},$$

whence

$$(\cot \phi + \cot \gamma)^2 = \cot^2 \gamma + \cot \alpha \cot \beta - 2.$$

This quantity is symmetric in α and β , so we should expect $\cot \angle UJS$ to be a similar expression and hence the two quantities to be negatives of each other. The details are left to the reader. $\square$

4 Some more practice problems

4.1 Mild Problems

Problem 16 (Caucasus 2022). *Let ABC be an isosceles triangle with $AC = BC$ and circumcircle k . The point D lies on the shorter arc of k over the chord BC and is different from B and C . Let E denote the intersection of CD and AB . Prove that the line through B and C is a tangent of the circumcircle of the triangle BDE .*

Problem 17 (Chile 2022). *Let ABC be a triangle such that $\angle CAB = 60^\circ$. Consider D, E points on sides AC and AB respectively such that BD bisects angle $\angle ABC$, CE bisects angle $\angle BCA$ and let I be the intersection of them. Prove that $|ID| = |IE|$.*

Problem 18 (Bangladesh 2022). *In $\triangle ABC$, $\angle BAC$ is a right angle. BP and CQ are bisectors of $\angle B$ and $\angle C$ respectively, which intersect AC and AB at P and Q respectively. Two perpendicular segments PM and QN are drawn on BC from P and Q respectively. Find the value of $\angle MAN$ with proof.*

Problem 19 (Canada 2011). *Let $ABCD$ be a cyclic quadrilateral with opposite sides not parallel. Let X and Y be the intersections of AB, CD and AD, BC respectively. Let the angle bisector of $\angle AXD$ intersect AD, BC at E, F respectively, and let the angle bisectors of $\angle AYB$ intersect AB, CD at G, H respectively. Prove that $EFGH$ is a parallelogram.*

4.2 Medium Problems

Problem 20 (ISL 2012). *Let ABC be a triangle with $AB \neq AC$ and circumcenter O . The bisector of $\angle BAC$ intersects BC at D . Let E be the reflection of D with respect to the midpoint of BC . The lines through D and E perpendicular to BC intersect the lines AO and AD at X and Y respectively. Prove that the quadrilateral $BXCY$ is cyclic.*

Problem 21 (ISL 2012). Let ABC be a triangle with $\angle BCA = 90^\circ$, and let D be the foot of the altitude from C . Let X be a point in the interior of the segment CD . Let K be the point on the segment AX such that $BK = BC$. Similarly, let L be the point on the segment BX such that $AL = AC$. Let M be the point of intersection of AL and BK . Show that $MK = ML$.

Problem 22 (ISL 2021). Let $ABCD$ be a quadrilateral inscribed in a circle Ω . Let the tangent to Ω at D meet rays BA and BC at E and F , respectively. A point T is chosen inside $\triangle ABC$ so that $\overline{TE} \parallel \overline{CD}$ and $\overline{TF} \parallel \overline{AD}$. Let $K \neq D$ be a point on segment DF satisfying $TD = TK$. Prove that lines AC , DT , and BK are concurrent.

4.3 Spicy Problems

Problem 23 (ISL 2017). A convex quadrilateral $ABCD$ has an inscribed circle with center I . Let I_a, I_b, I_c and I_d be the incenters of the triangles DAB, ABC, BCD and CDA , respectively. Suppose that the common external tangents of the circles AI_bI_d and CI_bI_d meet at X , and the common external tangents of the circles BI_aI_c and DI_aI_c meet at Y . Prove that $\angle XIY = 90^\circ$.

Problem 24 (IMO 2014). Convex quadrilateral $ABCD$ has $\angle ABC = \angle CDA = 90^\circ$. Point H is the foot of the perpendicular from A to BD . Points S and T lie on sides AB and AD , respectively, such that H lies inside triangle SCT and

$$\angle CHS - \angle CSB = 90^\circ, \quad \angle THC - \angle DTC = 90^\circ.$$

Prove that line BD is tangent to the circumcircle of triangle TSH .

Problem 25 (ISL 2014). Let ABC be a triangle with circumcircle Ω and incentre I . Let the line passing through I and perpendicular to CI intersect the segment BC and the arc BC (not containing A) of Ω at points U and V , respectively. Let the line passing through U and parallel to AI intersect AV at X , and let the line passing through V and parallel to AI intersect AB at Y . Let W and Z be the midpoints of AX and BC , respectively. Prove that if the points I, X , and Y are collinear, then the points I, W , and Z are also collinear.